PRINCIPLES OF ECONOMICS

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FPEC 101 – LECTURE 3

Unit 1

Demand and Supply Analysis

Outline

- Demand
- Supply
- Shifts
- Equilibrium

Market

- The price mechanism refers to the determination of price through the market forces of demand and supply. Prices act as signals for producers to show them what to produce and how to use their scarce resources. Resources are diverted to the production of those goods which are in demand as they will earn a greater profit.
- A market in the true economic sense is any effective arrangement where buyers and sellers come in contact with one another for the purpose of making transactions.

Demand

- **Demand** is the quantity of a product that consumers are
  - Willing to buy
  - Able to buy
  - At a price
  - Over a period of time

The Law of Demand – inverse relation between price and Qd.

- When goods are cheap, people buy more
- When Price falls , quantity demanded increases
- When goods are expensive, People buy less
- When Price rises , quantity demanded falls

Law of Demand

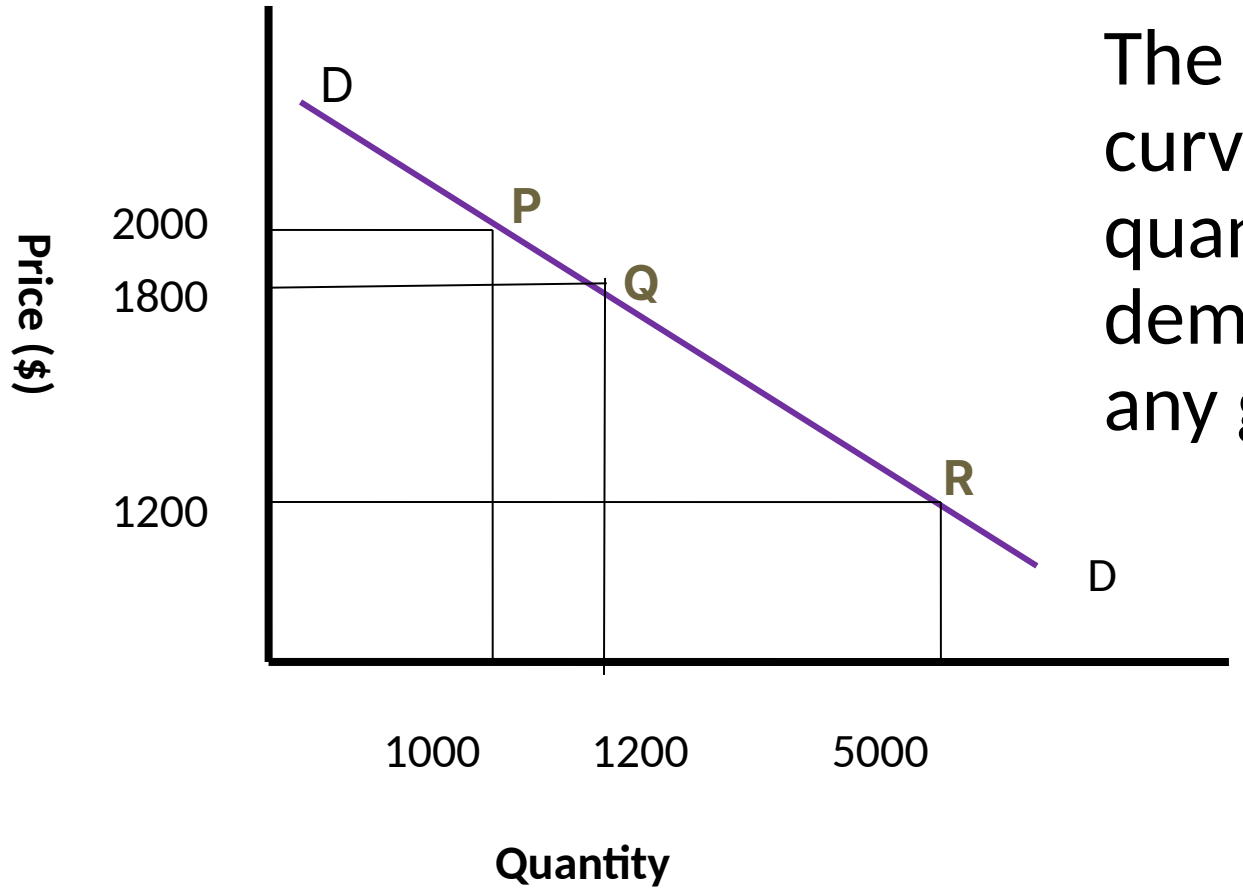
- “The Law of demand states that price and quantity demanded have a negative or inverse relationship. As price increases, quantity demanded decreases and vice Versa.” Ceteris paribus**.**
- Consumption is subject to *diminishing marginal utility.* And because successive units of a particular product yield less and less marginal utility, consumers will buy additional units only if the price of those units is progressively reduced.
- The *income effect* indicates that a lower price increases the purchasing power of a buyer’s money income, enabling the buyer to purchase more of the product than before. A higher price has the opposite effect.
- The *substitution effect* suggests that at a lower price buyers have the incentive to substitute what is now a less expensive product for similar products that are now relatively more expensive. The product whose price has fallen is now “a better deal” relative to the other products.

Demand schedule

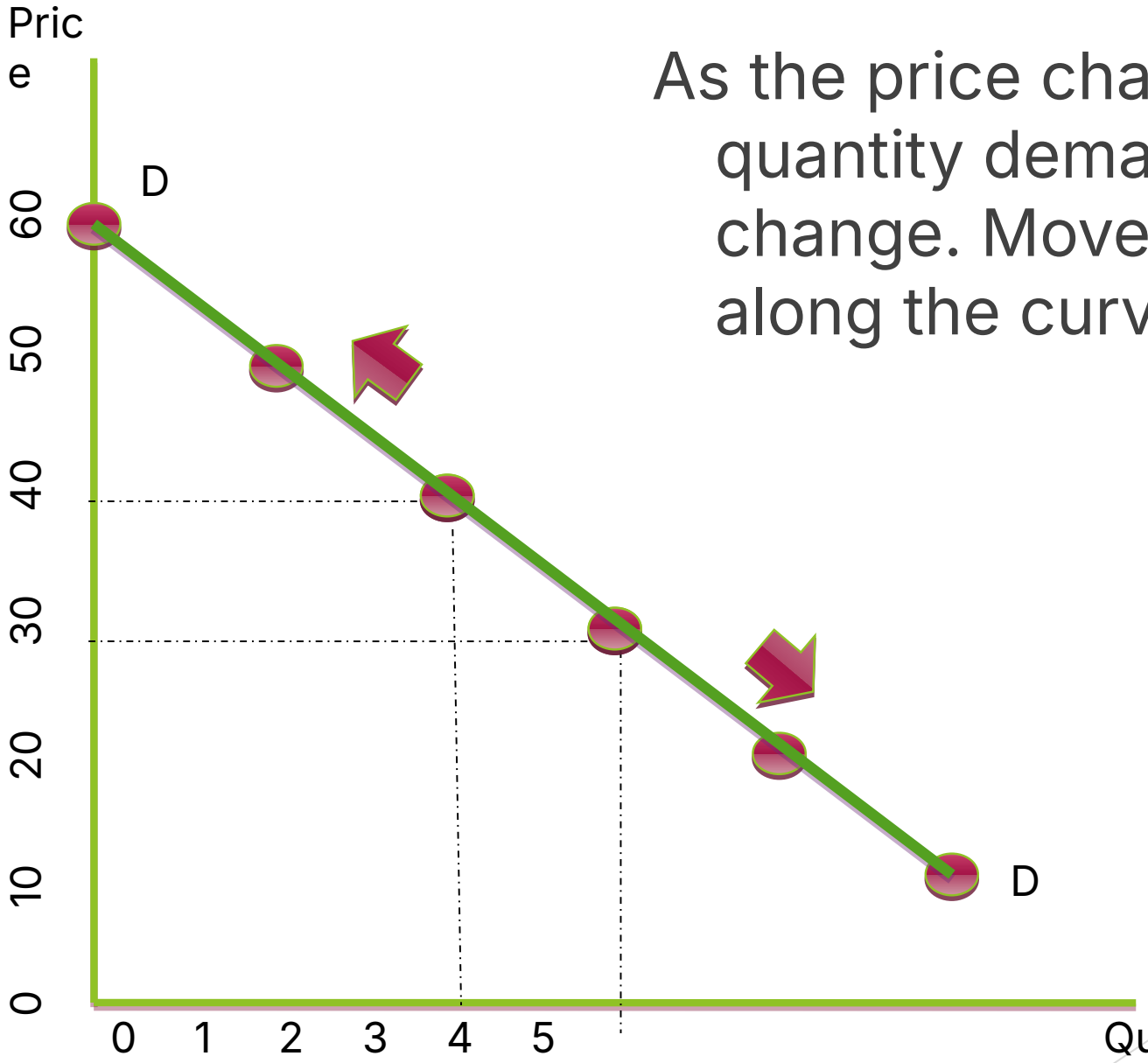
Price of computers	Quantity demanded per week
2000	1000
1800	2000
1600	3000
1400	4000
1200	5000

The Demand Schedule shows quantities demanded at given price.

The Demand Curve



The demand curve shows the quantity demanded at any given price.



As the price changes, the quantity demanded will change. Movement along the curve.

Individual Demand and Market Demand

- *Individual demand* is the demand of just one consumer and is referred to as *quantity demanded* while the total demand for a product from all its consumers is known as *market demand*.
- An important assumption that we make when discussing the factors that affect the demand is the **ceteris paribus** assumption. Under this assumption, we only discuss a change in one factor at a time.

Individual Demand and Market Demand

TABLE 3.1 Market Demand for Corn, Three Buyers

Price per Bushel	Quantity Demanded						Total Quantity Demanded per Week
	Joe		Jen		Jay		
$5	10	+	12	+	8	=	30
4	20	+	23	+	17	=	60
3	35	+	39	+	26	=	100
2	55	+	60	+	39	=	154
1	80	+	87	+	54	=	221

Economics by Mc Connell & Brue pg 47

<u>Difference between quantity demanded and demand</u>

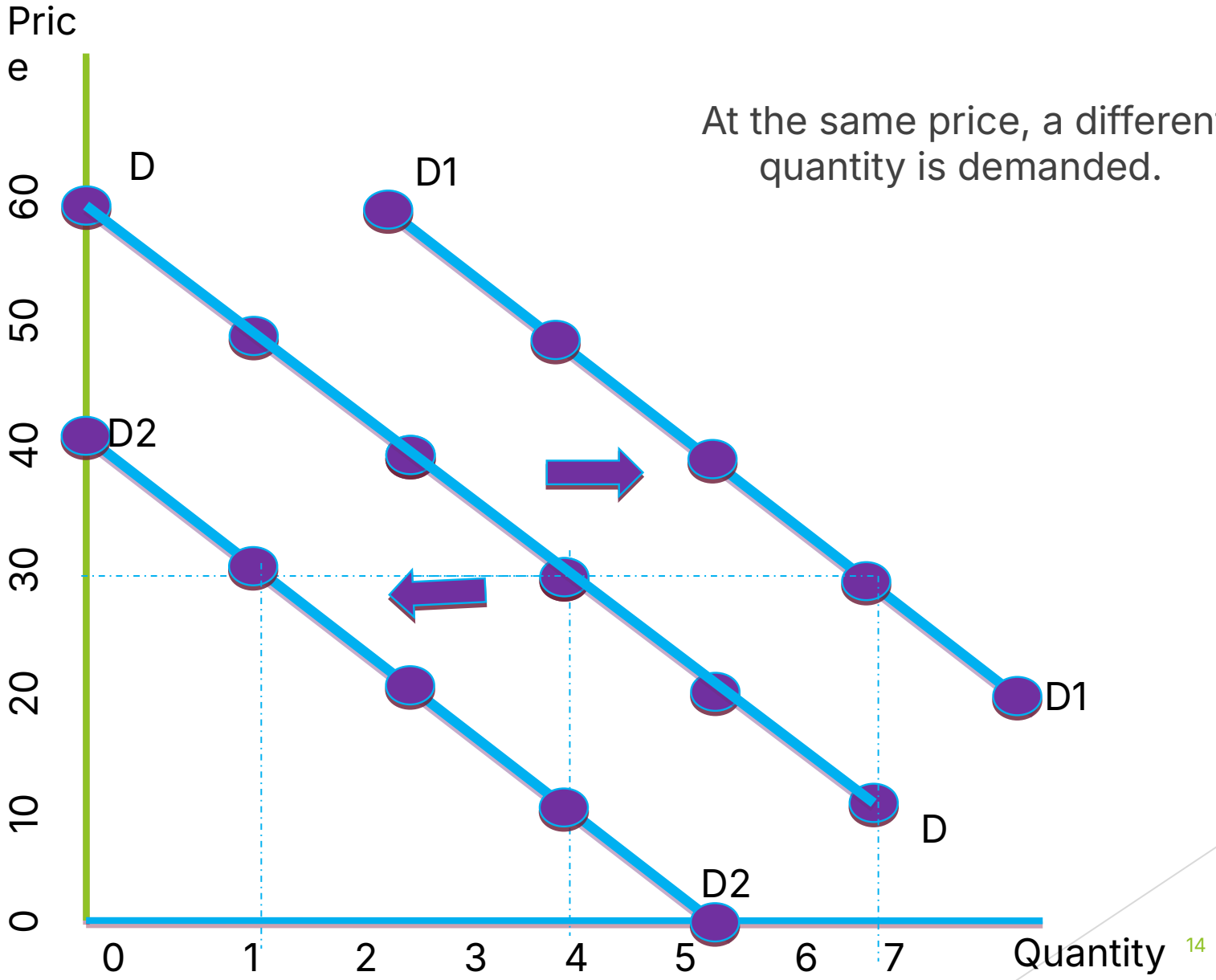
A change in price brings about a change in quantity demanded and is a movement along the curve.

A change in any other factor besides price will shift the curve so that more or less is demanded at the same price level. A change in price **never** brings about a movement or shift of the curve.

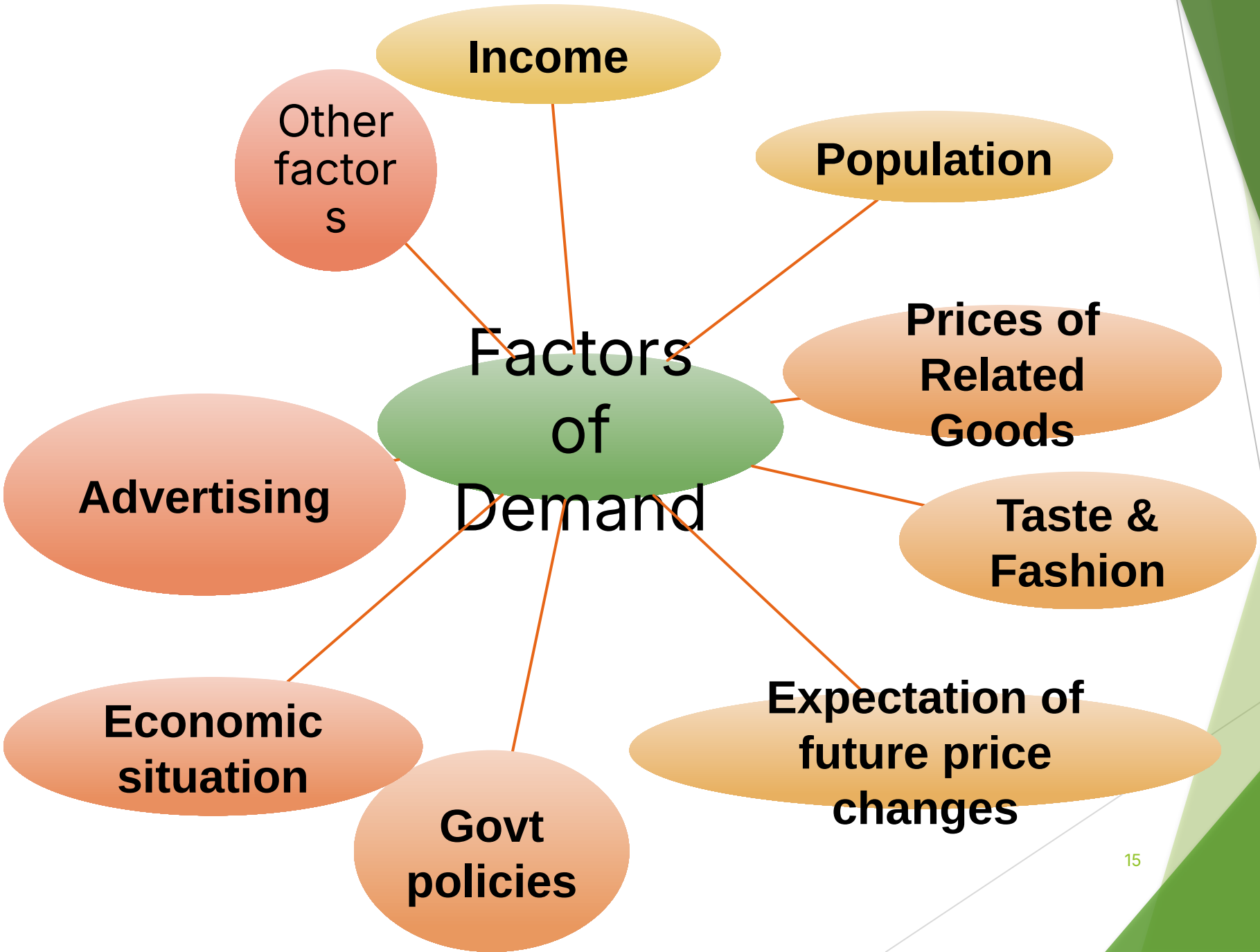
Demand - Shift

Shift in the demand curve

- A change in any factor ***besides price*** will bring about a parallel shift, or movement of the entire demand curve. This means that more or less is demanded at each and every price level than they did before.
- Factors affecting demand, conditions of demand, determinants of demand



At the same price, a different quantity is demanded.

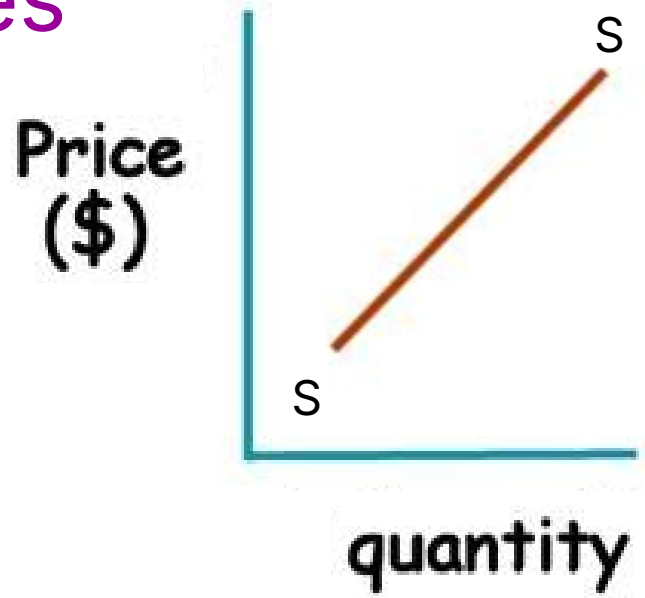

Income
Other factors
Population
Factors of Demand
Prices of Related Goods
Advertising
Taste & Fashion
Economic situation
Expectation of future price changes
Govt policies

LAW OF SUPPLY

- **“This is the producers willingness and ability to sell a product at a given price over a given period of time .”** It refers to the amount producers are willing to create and sell at a given price level. Existing stocks are not considered supply unless they are put up for sale._
- A firm interested in profit will only make and sell a product if it can do so at a price over and above what it cost the firm to make. Hence, the higher the price, the more the firm is willing to supply because it expects more profits.
- ***The Law of supply states that price and quantity supplied have a positive or direct relation. As price increases quantity supplied increases and vice Versa.”***

LAW OF SUPPLY

- Price increases, quantity Supplied increases
- Price decreases, quantity Supplied decreases



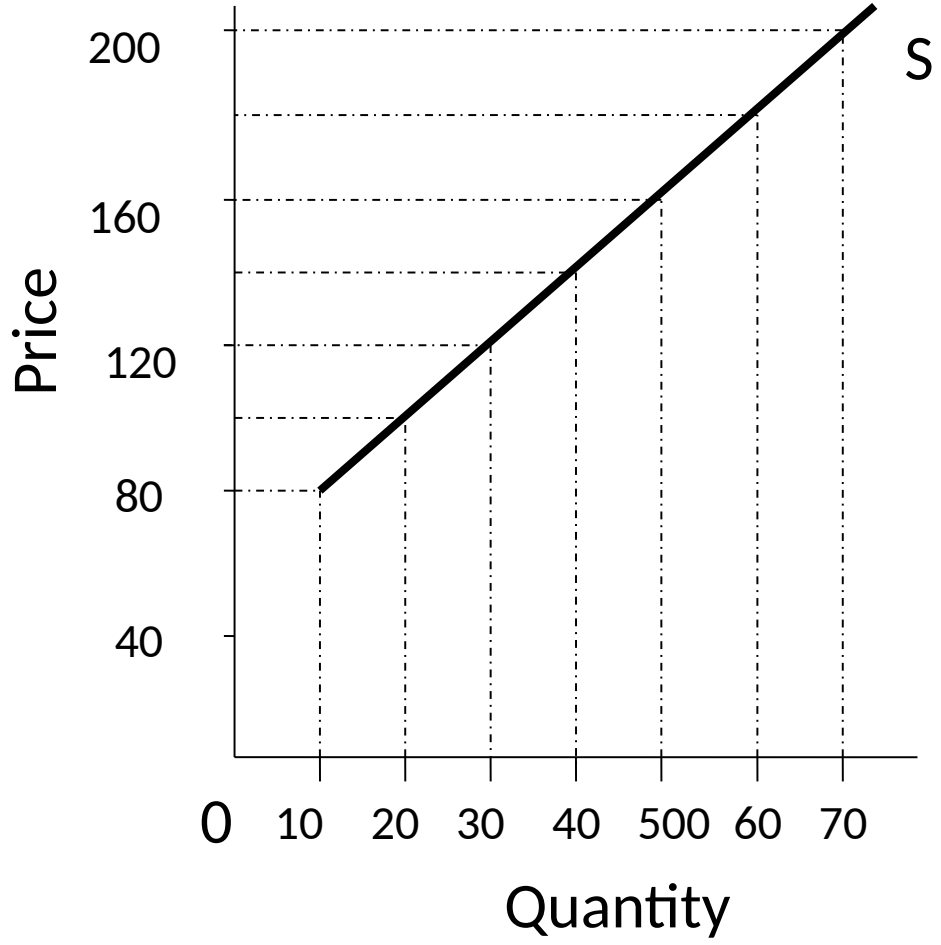
SUPPLY

- **Individual Supply** - the supply of one Firm/ Producer
- **Market Supply** is the total Supply of the Market

- **<u>Difference between quantity Supplied and Supply</u>**

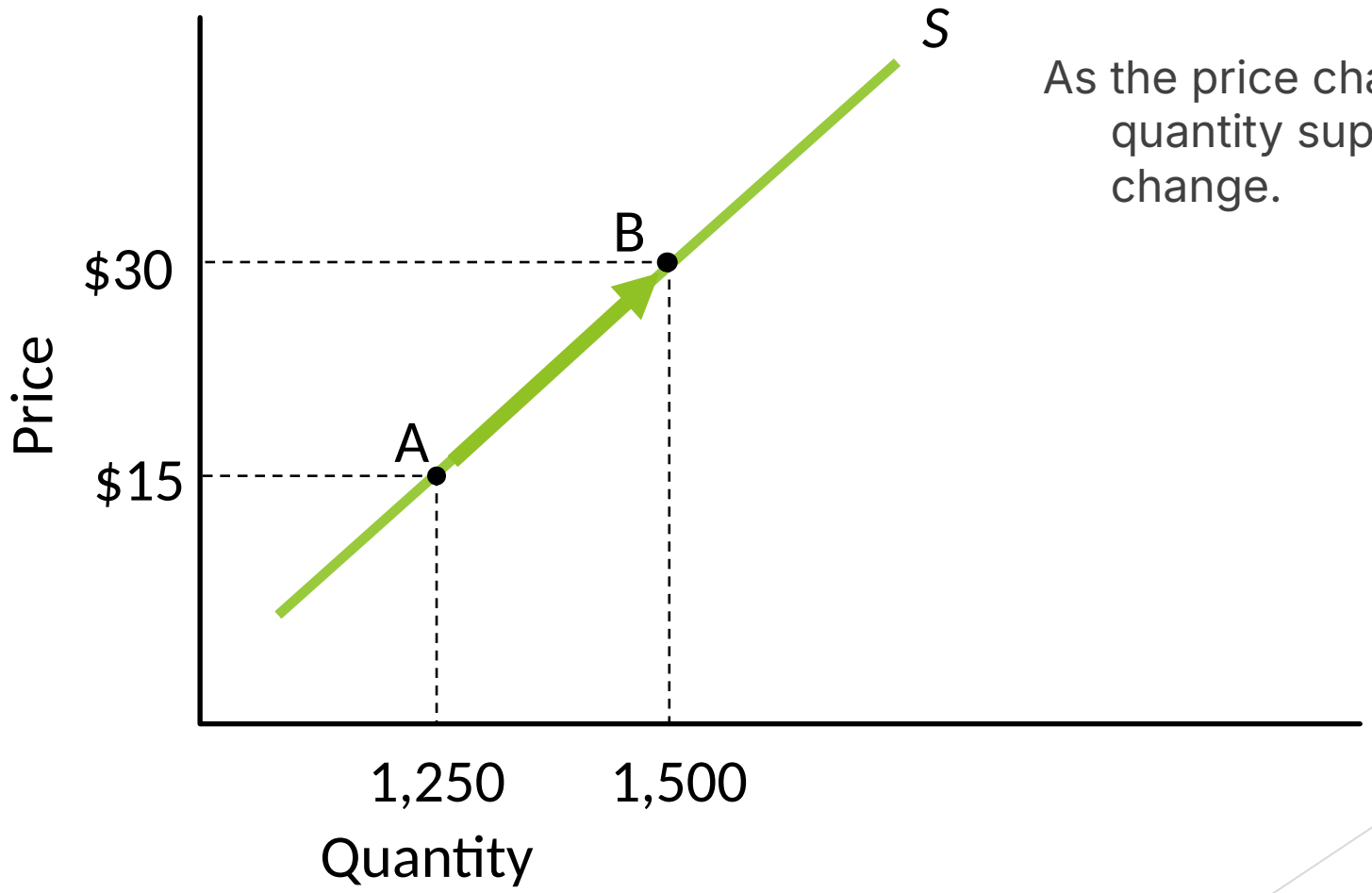
Just as we discussed for demand, it is important to remember that a change in quantity supplied is brought about by a change in price and is a movement along the curve. Any other factor changing besides a change in price brings about a shift of the curve. A change in price *never* brings about a movement or shift of the curve

SUPPLY



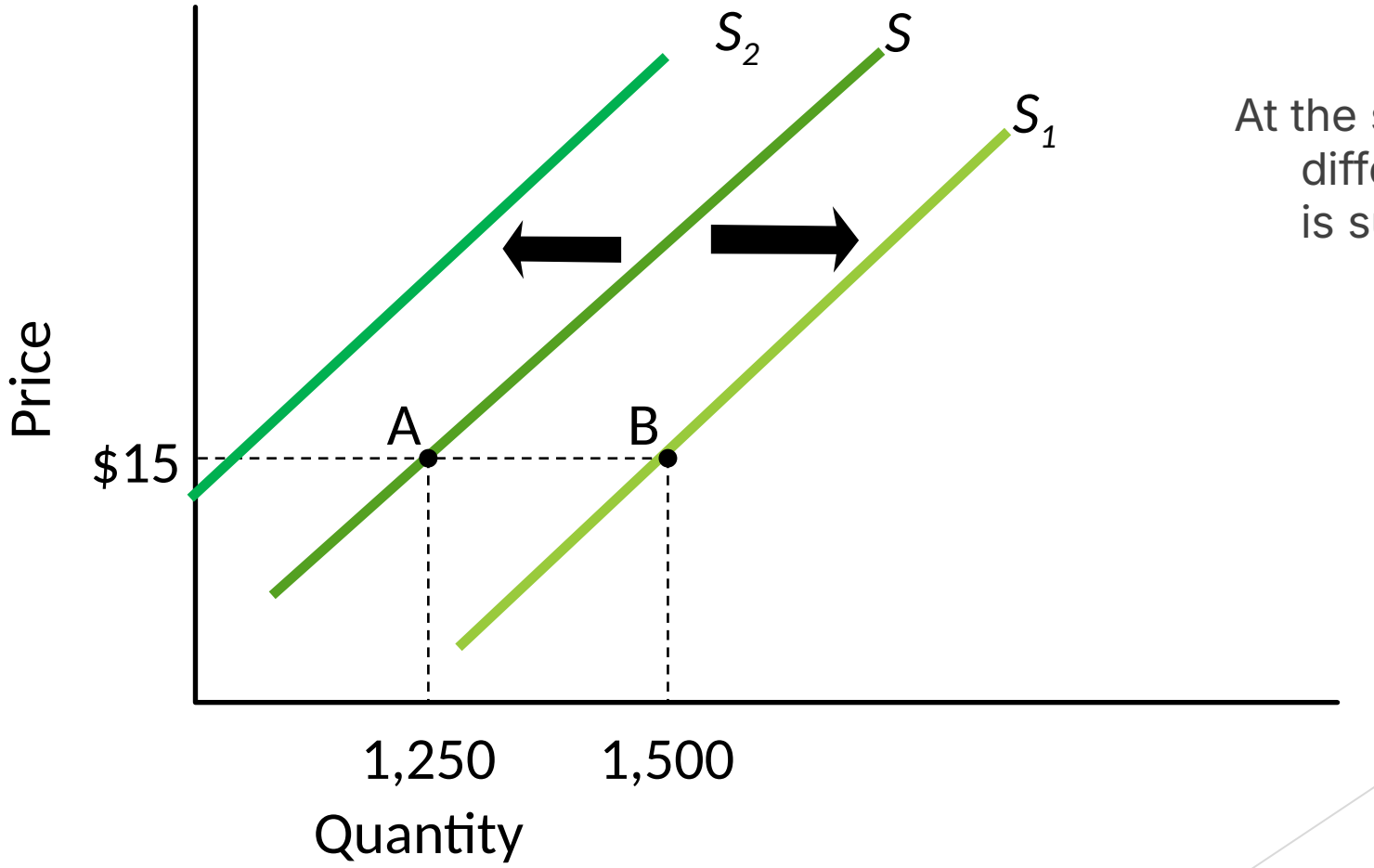
The Supply curve shows the quantity supplied at any given price.

Movement along Supply Curve

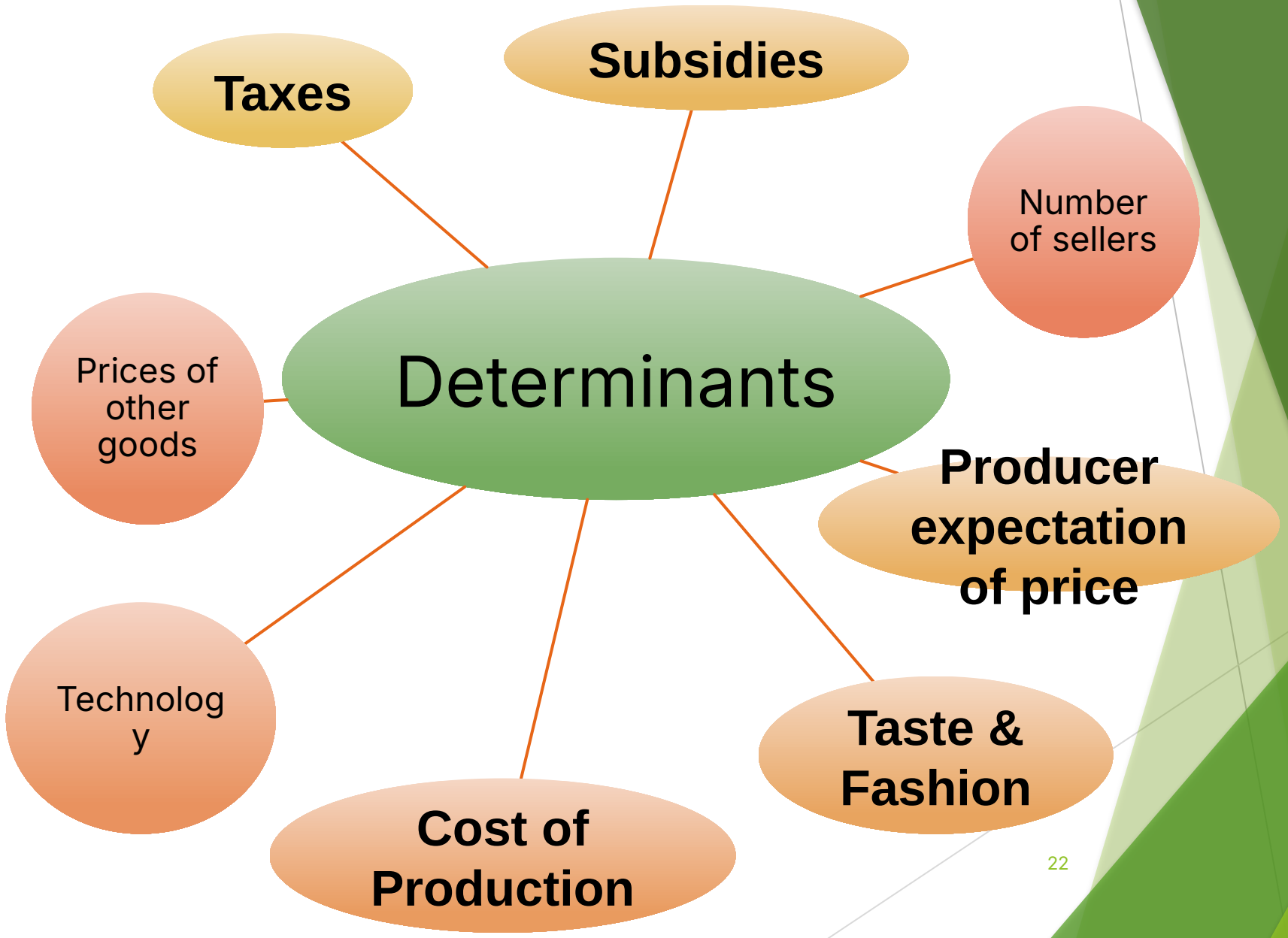


As the price changes, the quantity supplied will also change.

Shift of Supply Curve



At the same price, a different quantity is supplied.


Taxes
Subsidies
Number of sellers
Prices of other goods
Determinants
Producer expectation of price
Technology
Taste & Fashion
Cost of Production

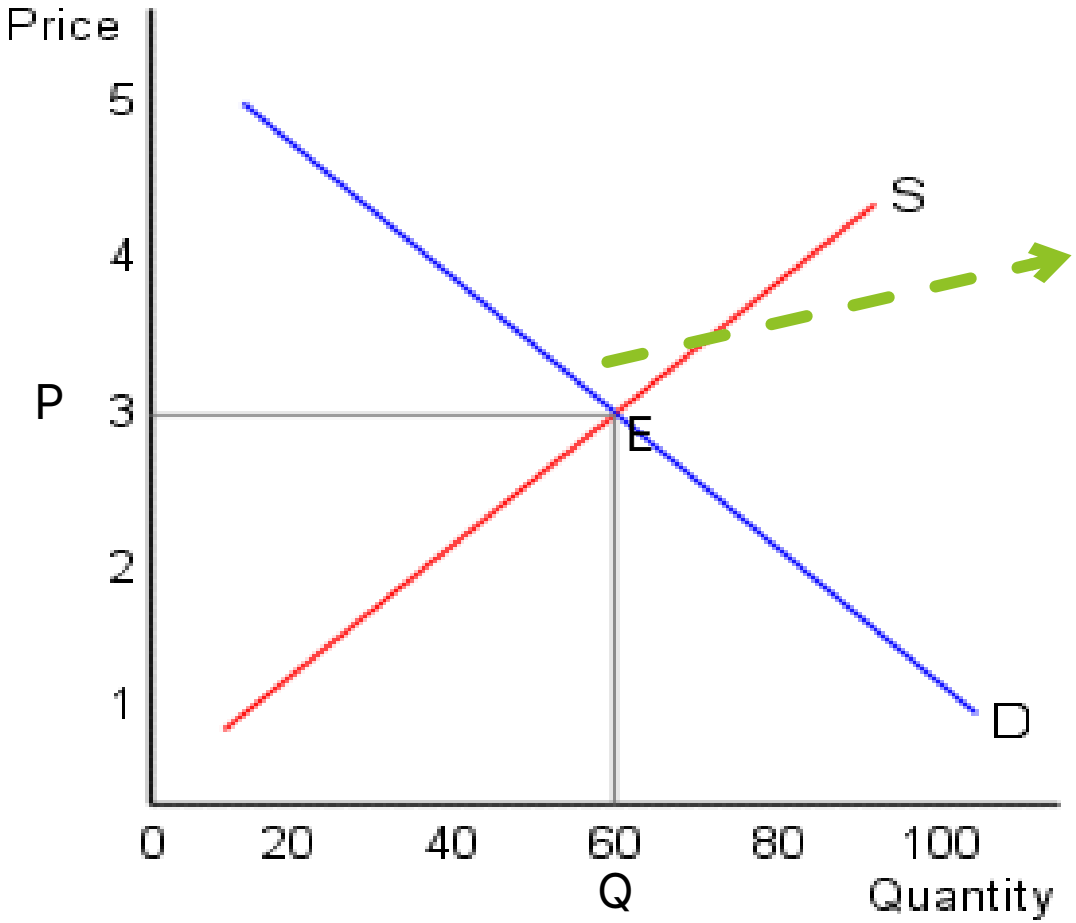
Functions of Price Mechanism

- Allocate: The market mechanism allocates scarce resources.
- Ration : price serve to ration scarce resources when demand outstrips supply.
- Signal : Market prices adjust to demonstrate where resources are required, and when they are not.
- Incentive : When the price of a product rises, the quantity supplied increases, this is due to the incentive function.

EQUILIBRIUM

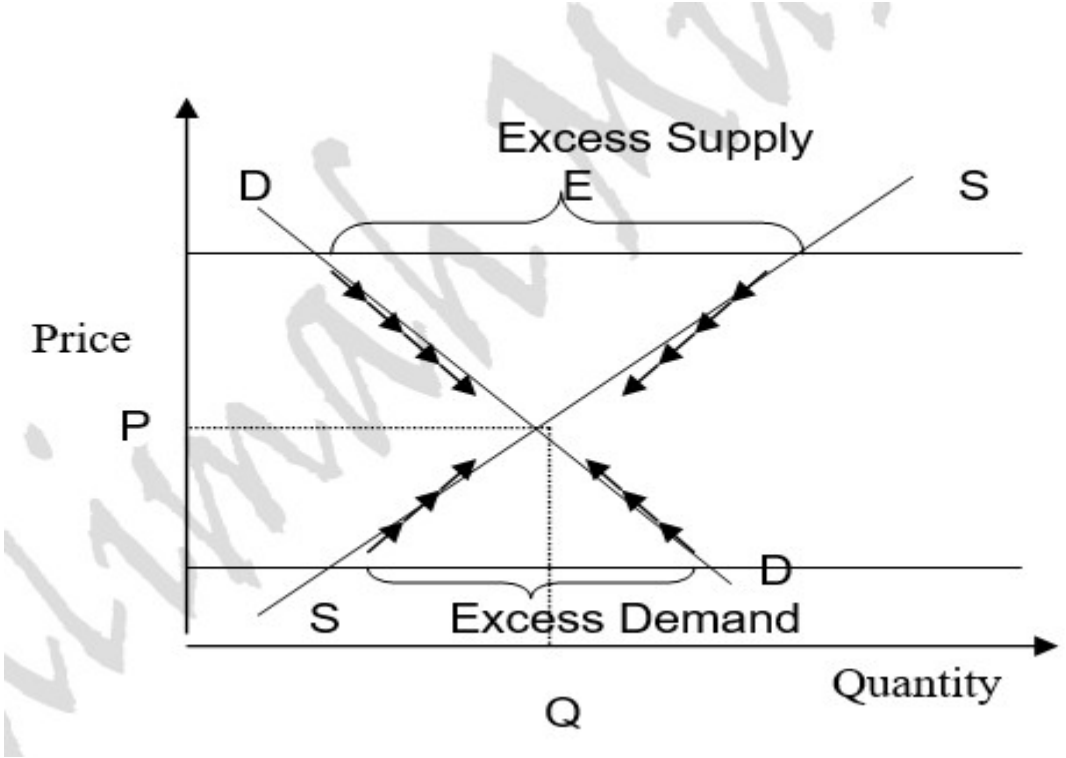
- Demand = Supply → **Equilibrium**
- Demand ≠ Supply → **Disequilibrium**
  - Surplus = Supply > Demand
  - Deficit = Demand > Supply
- The point of intersection shows the equilibrium point ‘E’. At this point **D=S=Qd=Qs=Q**
- It is also known as the market clearing price. At any other point the market will be in disequilibrium. This is the point where the market forces will come to rest and there is no tendency to change.

EQUILIBRIUM



Demand = Supply (Equilibrium Point)

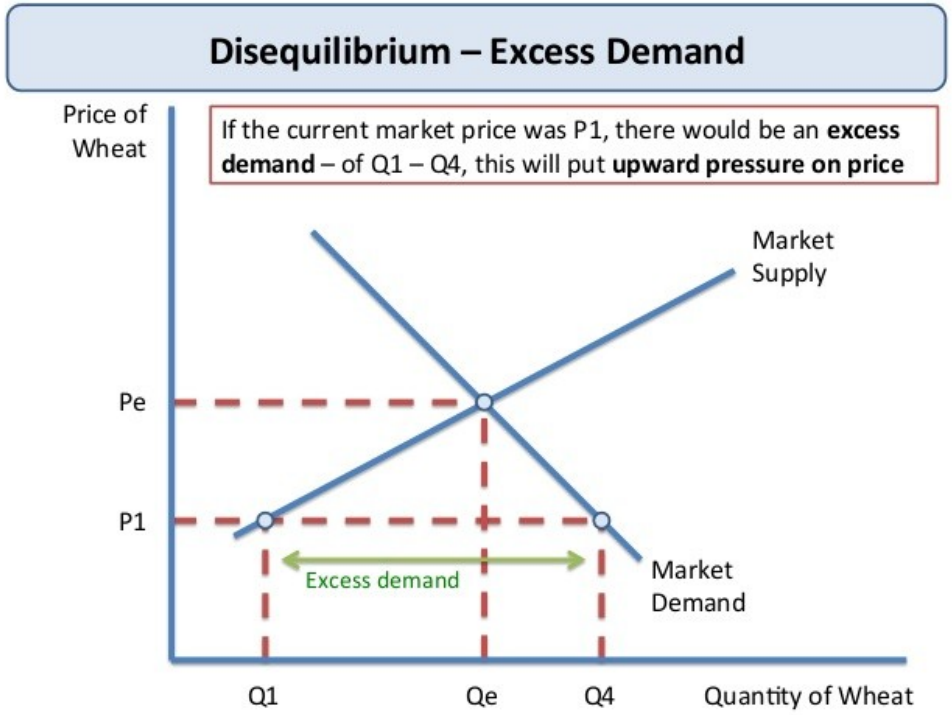
DISEQUILIBRIUM



MARKET FAILURE AND GOVERNMENT INTERVENTION

DISEQUILIBRIUM – EXCESS DEMAND

- A ***price ceiling*** sets the maximum legal price a seller may charge for a product or service. A price at or below the ceiling is legal; a price above it is not.
- When the ceiling price is below the equilibrium price, a persistent product shortage results.

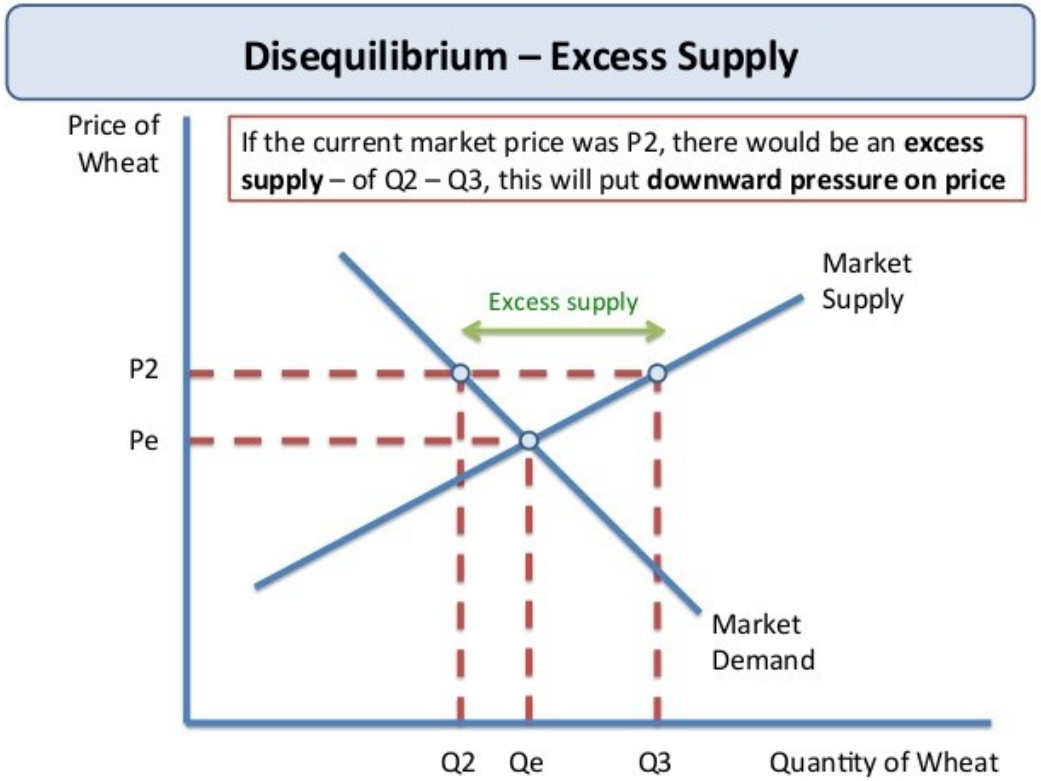


DISEQUILIBRIUM- ED / SHORTAGE

- Price ceilings are imposed by government on specific products, mostly necessities, to enable consumers to afford. Examples are rent controls and usury laws, which specify maximum “prices” in the forms of rent and interest that can be charged to borrowers.
- Rationing Problem
  Issuing ration coupons authorizing bearers to purchase a fixed amount of product monthly. These are then equitably distributed so that the wealthy family of four and the poor family of four both receive the same number of coupons.
- Black Markets
  Ration coupons may give rise to black marketing as many buyers are willing to pay more than the ceiling price and that is more profitable for the suppliers. A sizable enforcement bureaucracy would be required to ensure price controls.
  Counterfeiting of ration coupons will be a problem.
  Since government sets price, it might face political pressure to set an even lower price.
  Example ceiling on rent. It may result in demand to increase but inadequate supply.
- Resource Misallocation
  In brief, controls distort market signals and thus *resources are misallocated.* Too few resources are allocated and too many alternative uses, eventually being the primary cause of such shortages.

DISEQUILIBRIUM- EXCESS SUPPLY

A price floor is a minimum price fixed by the government. A price at or above the price floor is legal; a price below it is not. Usually for agricultural products. When the price *floor* is above the equilibrium price, a persistent product surplus results.



DISEQUILIBRIUM – ES/ SURPLUS

- Price floors above equilibrium prices are usually invoked when society feels that the free functioning of the market system has not provided a sufficient income for certain groups of resource suppliers or producers. Supported prices for agricultural products and current minimum wages are two examples of price (or wage) floors
- The government may cope with the surplus resulting from a price floor in two ways
  Supply restriction
  It can restrict supply or increase demand through alternative uses for the product, thus reducing the surplus.
  Purchase excess stock
  Government must purchase the surplus output at the adjusted price and store or otherwise dispose of it

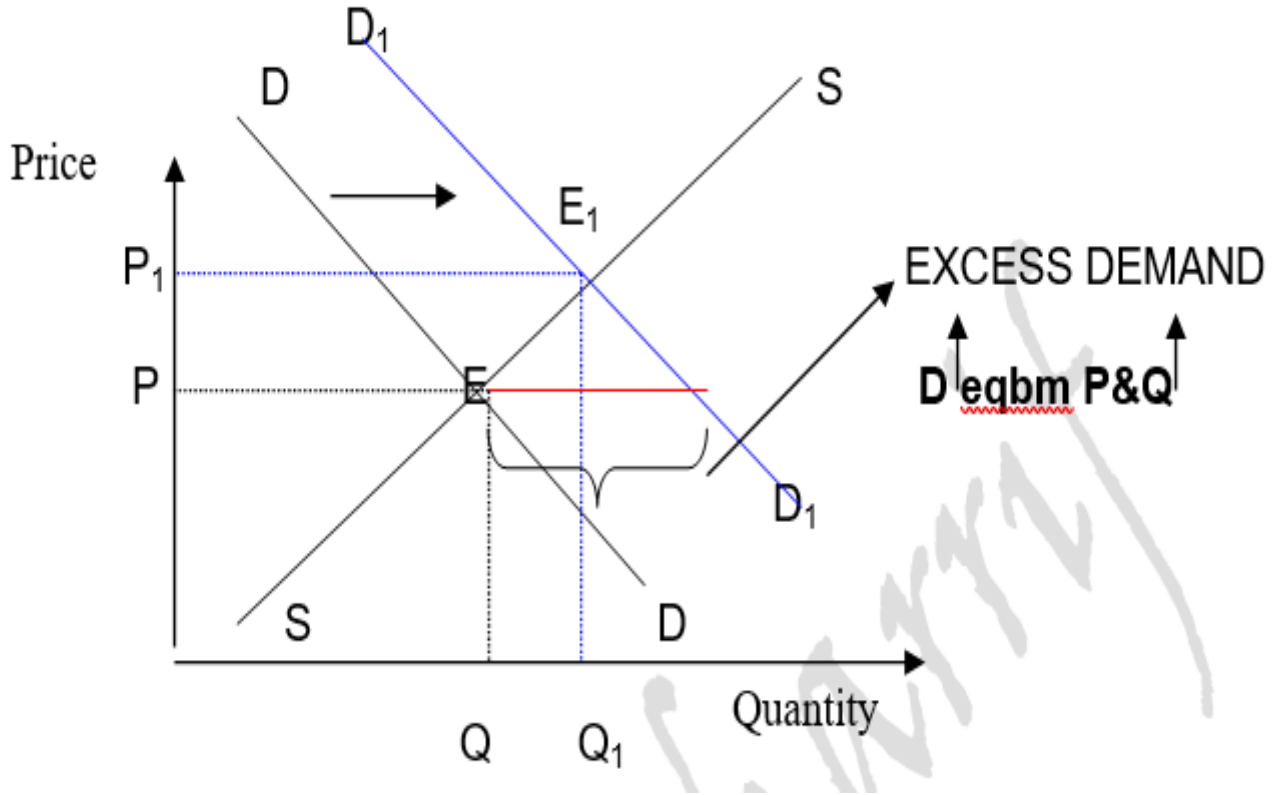
DISEQUILIBRIUM – ES/ SURPLUS

- Additional Consequences
  Price floors also disrupt the rationing ability of prices as well as *resource allocation.* Without price floor producer would face losses and force high-cost wheat producers to plant other crops or abandon farming altogether. So society devotes too many of its scarce resources to wheat production and too few to producing other, more valuable, goods and services. It fails to achieve *allocative efficiency.*
- Consumers of wheat-based products pay higher prices because of the price floor.
- Taxpayers pay higher taxes to finance the government's purchase of the surplus.
- Price floor causes potential environmental damage through use of less suitable resources.
- The higher price encourages imports but since such imports would increase the quantity of wheat supplied and thus undermine the price floor, the government needs to erect tariffs (taxes on imports) to keep the foreign wheat out. Such tariffs usually prompt other countries to retaliate.
- Government controlled prices cause shortages or surpluses, distort resource allocation, and produce negative side effects.

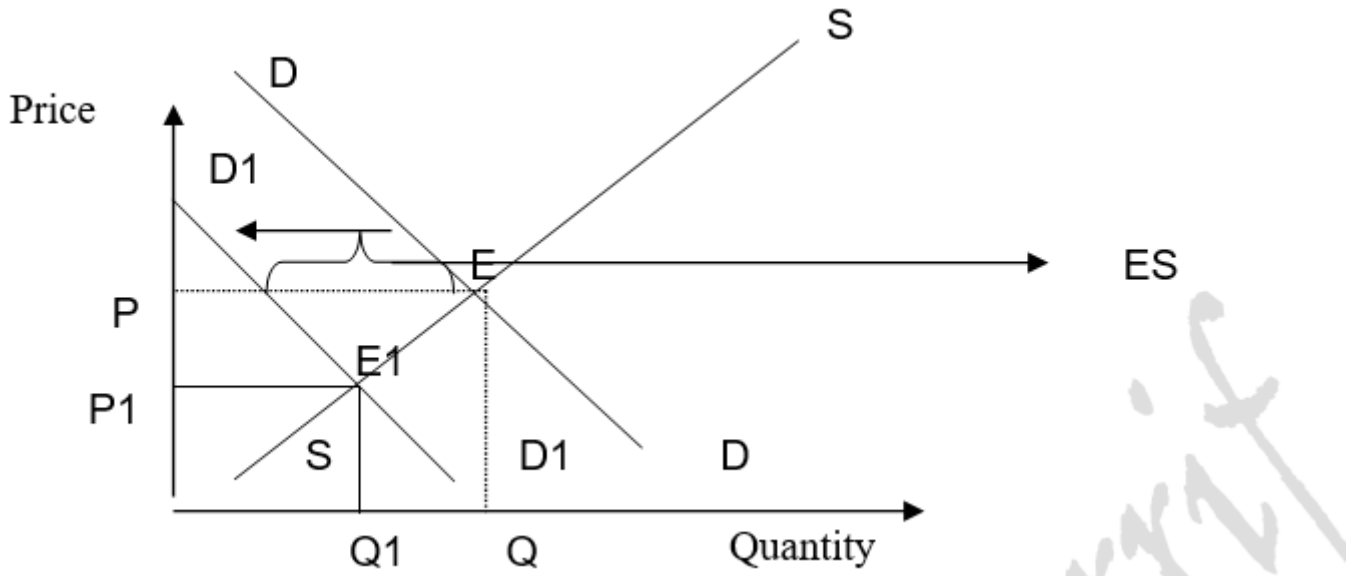
Why the Equilibrium Changes?

Change in Demand
Change in Supply
Change in Demand & Supply

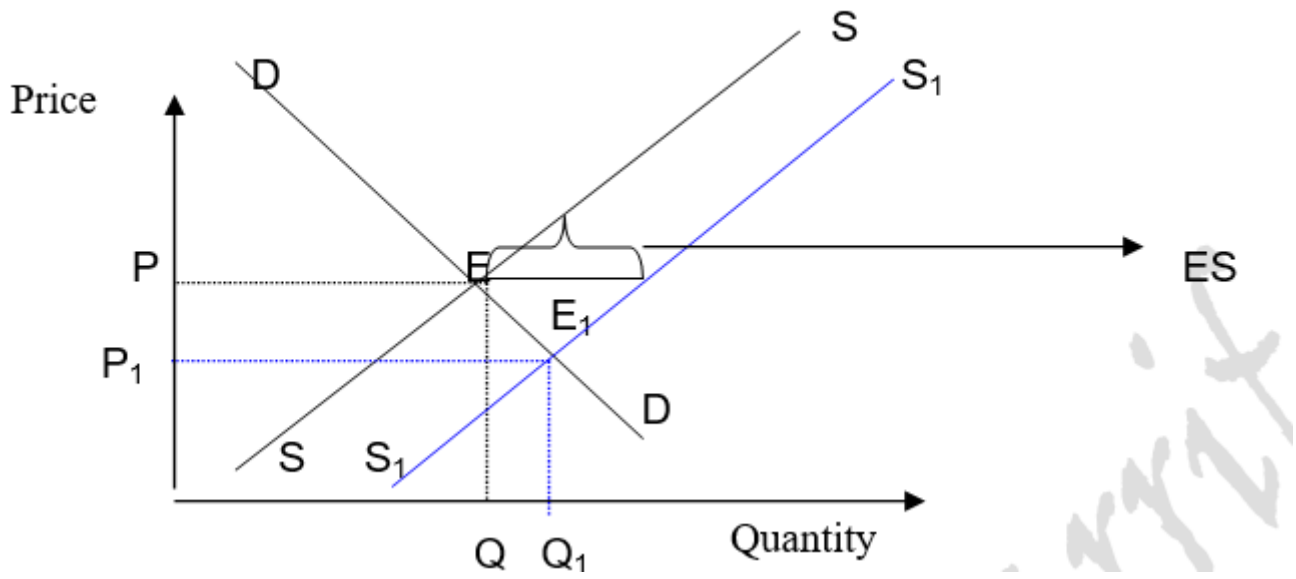
SHIFT IN DEMAND - INCREASE



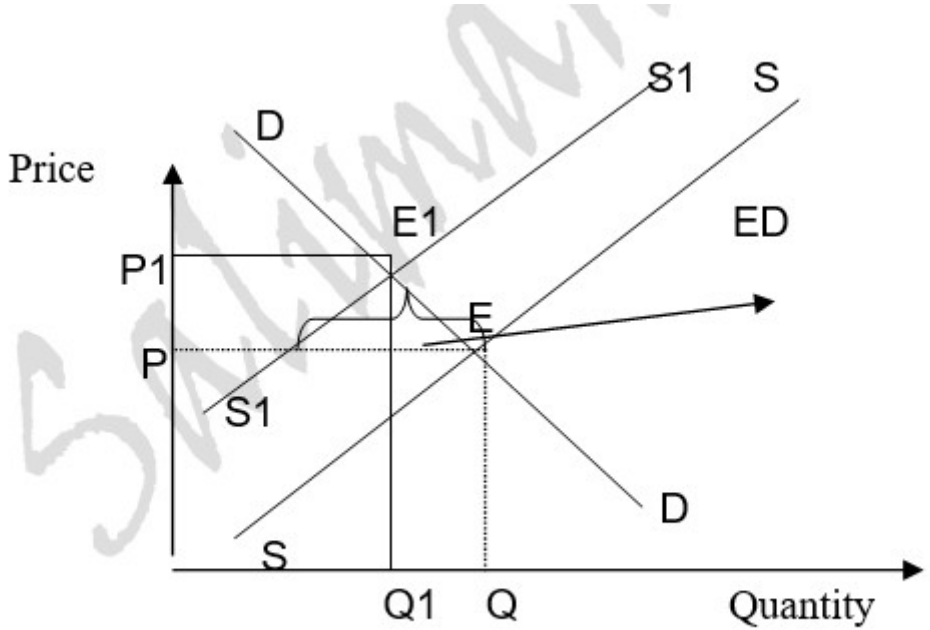
SHIFT IN DEMAND - DECREASE



SHIFT IN SUPPLY - INCREASE



SHIFT IN SUPPLY - DECREASE



Outline

- Demand
- Supply
- Shifts
- Equilibrium
- Disequlibrium